AI/ML Software Engineer

Job Family: Software Development & Support

Travel Required: None

Clearance Required: Ability to Obtain Public Trust

What You Will Do

Guidehouse Technology is seeking a highly skilled and motivated AI-focused Software Engineer to join our dynamic team. In this role, you will work with federal clients to design, develop, and deploy innovative Artificial Intelligence and Machine Learning solutions that address immediate mission challenges. You will be a key contributor on a team of data scientists, developers, and subject matter experts, applying best practices to build robust AI applications, sophisticated data pipelines, and intelligent systems.

The ideal candidate is a fast learner and rapid developer with a strong foundation in software engineering and demonstrable experience building production-ready AI solutions from concept to deployment. The ideal candidate should also possess strong hands-on expertise in Python, machine learning, graph analytics, natural language processing, and entity resolution. This combination of skills is essential for uncovering hidden patterns across large, disparate datasets in support of fraud detection, trafficking detection, and other mission-critical investigative use cases.

A critical competency for this role is network analysis—the ability to link individuals, transactions, organizations, events, documents, and locations to identify relationships, detect anomalies, and expose organized illegal activities. In tandem, the candidate should be proficient in developing and applying AI/ML models for risk scoring, anomaly detection, pattern recognition, and investigative prioritization, enabling federal clients to enhance operational effectiveness and focus resources on the highest-risk cases.

What You Will Do

Design, develop, and deploy end-to-end AI/ML applications, with a focus on Retrieval-Augmented Generation systems, AI chatbots, and agentic workflows.
Construct and maintain scalable data pipelines for processing, transforming, linking, and feeding structured and unstructured data into AI models and applications.
Develop machine learning models for risk scoring, anomaly detection, classification, clustering, and predictive analytics to support investigative and operational decision-making.
Apply graph analytics and network analysis techniques to identify hidden relationships among individuals, transactions, locations, organizations, and events.
Implement entity resolution capabilities to match, deduplicate, and link records across large, disparate datasets.

Apply NLP techniques to extract entities, relationships, topics, events, and indicators from unstructured text, documents, case notes, and other mission-relevant data sources.
Collaborate with cross-functional teams to rapidly prototype and iterate on solutions for mission-critical challenges in areas such as fraud detection, trafficking detection, data triage, search and optimization, and automated discovery.
Develop and maintain REST APIs to serve model inferences and integrate AI capabilities into larger systems.
Contribute to the team's software architecture design, ensuring solutions are scalable, reliable, secure, and efficient.
Deploy and manage applications and models in cloud environments, primarily AWS, leveraging infrastructure-as-code and DevOps best practices.
What You Will Need

Must be able to OBTAIN and MAINTAIN a Federal or DoD "PUBLIC TRUST"; candidates must obtain approved adjudication of their PUBLIC TRUST prior to onboarding with Guidehouse.
Candidates with an ACTIVE PUBLIC TRUST or SUITABILITY are preferred.
Bachelor’s degree is required.
Minimum SIX (6) years of hands-on experience with Python and JavaScript and/or TypeScript. A Master’s degree OR Industry Certification may be substituted for up to two years of relevant professional experience.
Strong hands-on experience with Python for AI/ML application development, data engineering, data analysis, and model implementation.
Proven experience developing RAG applications, including implementing re-ranking strategies.
Experience building AI chatbots or conversational agents.
Hands-on experience with AI application frameworks such as LangChain, Haystack, crewAI, or similar.
Strong knowledge of core Python data science and ML libraries, including NumPy, Pandas, Scikit-learn, NLTK, and OpenCV.
Demonstrated experience developing and applying machine learning models for classification, clustering, risk scoring, anomaly detection, and pattern recognition.
Experience with graph analytics, network analysis, and relationship discovery across complex datasets.
Experience with entity resolution, record linkage, deduplication, identity matching, or similar data matching techniques.
Experience applying NLP techniques such as named entity recognition, text classification, semantic search, information extraction, topic modeling, and relationship extraction.
Familiarity with deep learning frameworks such as PyTorch or TensorFlow.
Experience with search technologies such as Elasticsearch or OpenSearch.
Experience with relational databases, such as PostgreSQL or Oracle DB, and in-memory analytics databases, such as DuckDB.
Strong knowledge of SQL and data modeling techniques.
Experience with cloud SDKs, such as Boto3 for AWS.
Ability to work with large, disparate datasets and uncover hidden patterns, relationships, anomalies, and risk indicators.

Strong analytical, problem-solving, and communication skills, with the ability to translate mission needs into practical AI/ML solutions.
What Would Be Nice To Have

Familiarity with agentic AI frameworks such as AWS Strands Agents, PydanticAI, or similar.

Advanced prompt engineering skills for complex tasks beyond code generation.

Experience with asynchronous Python development.

Experience with MCP servers and tool calling within agentic workflows.

Knowledge of GPU-accelerated computing, CUDA, and hardware optimization for running ML models efficiently.

Experience with graph databases or graph processing frameworks such as Neo4j, Amazon Neptune, TigerGraph, NetworkX, GraphFrames, or similar.

Experience supporting fraud detection, trafficking detection, investigations, intelligence analysis, case management, or law enforcement-adjacent mission environments.

Experience building explainable AI/ML models that support analyst review, investigative workflows, and auditability.

Experience integrating AI/ML capabilities into federal cloud environments while following security, privacy, and compliance best practices.

Familiarity with geospatial analytics, temporal analytics, link analysis, or behavioral pattern detection.

Experience designing AI/ML solutions that support human-in-the-loop review, case prioritization, and investigative decision support. The annual salary range for this position is $113,000.00-$188,000.00. Compensation decisions depend on a wide range of factors, including but not limited to skill sets, experience and training, security clearances, licensure and certifications, and other business and organizational needs.

What We Offer:

Guidehouse offers a comprehensive, total rewards package that includes competitive compensation and a flexible benefits package that reflects our commitment to creating a diverse and supportive workplace.

Benefits include:

Medical, Rx, Dental & Vision Insurance
Personal and Family Sick Time & Company Paid Holidays
Parental Leave
401(k) Retirement Plan
Group Term Life and Travel Assistance
Voluntary Life and AD&D Insurance
Health Savings Account, Health Care & Dependent Care Flexible Spending Accounts
Transit and Parking Commuter Benefits
Short-Term & Long-Term Disability
Tuition Reimbursement, Personal Development, Certifications & Learning Opportunities
Employee Referral Program
Corporate Sponsored Events & Community Outreach
Care.com annual membership
Employee Assistance Program
Supplemental Benefits via Corestream (Critical Care, Hospital Indemnity, Accident Insurance, Legal Assistance and ID theft protection, etc.)
Position may be eligible for a discretionary variable incentive bonus
About Guidehouse

Guidehouse is an Equal Opportunity Employer–Protected Veterans, Individuals with Disabilities or any other basis protected by law, ordinance, or regulation.

Guidehouse will consider for employment qualified applicants with criminal histories in a manner consistent with the requirements of applicable law or ordinance including the Fair Chance Ordinance of Los Angeles and San Francisco.

If you have visited our website for information about employment opportunities, or to apply for a position, and you require an accommodation, please contact Guidehouse Recruiting at 1-571-633-1711 or via email at RecruitingAccommodation@guidehouse.com. All information you provide will be kept confidential and will be used only to the extent required to provide needed reasonable accommodation.

All communication regarding recruitment for a Guidehouse position will be sent from Guidehouse email domains including @guidehouse.com or guidehouse@myworkday.com. Correspondence received by an applicant from any other domain should be considered unauthorized and will not be honored by Guidehouse. Note that Guidehouse will never charge a fee or require a money transfer at any stage of the recruitment process and does not collect fees from educational institutions for participation in a recruitment event. Never provide your banking information to a third party purporting to need that information to proceed in the hiring process.

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